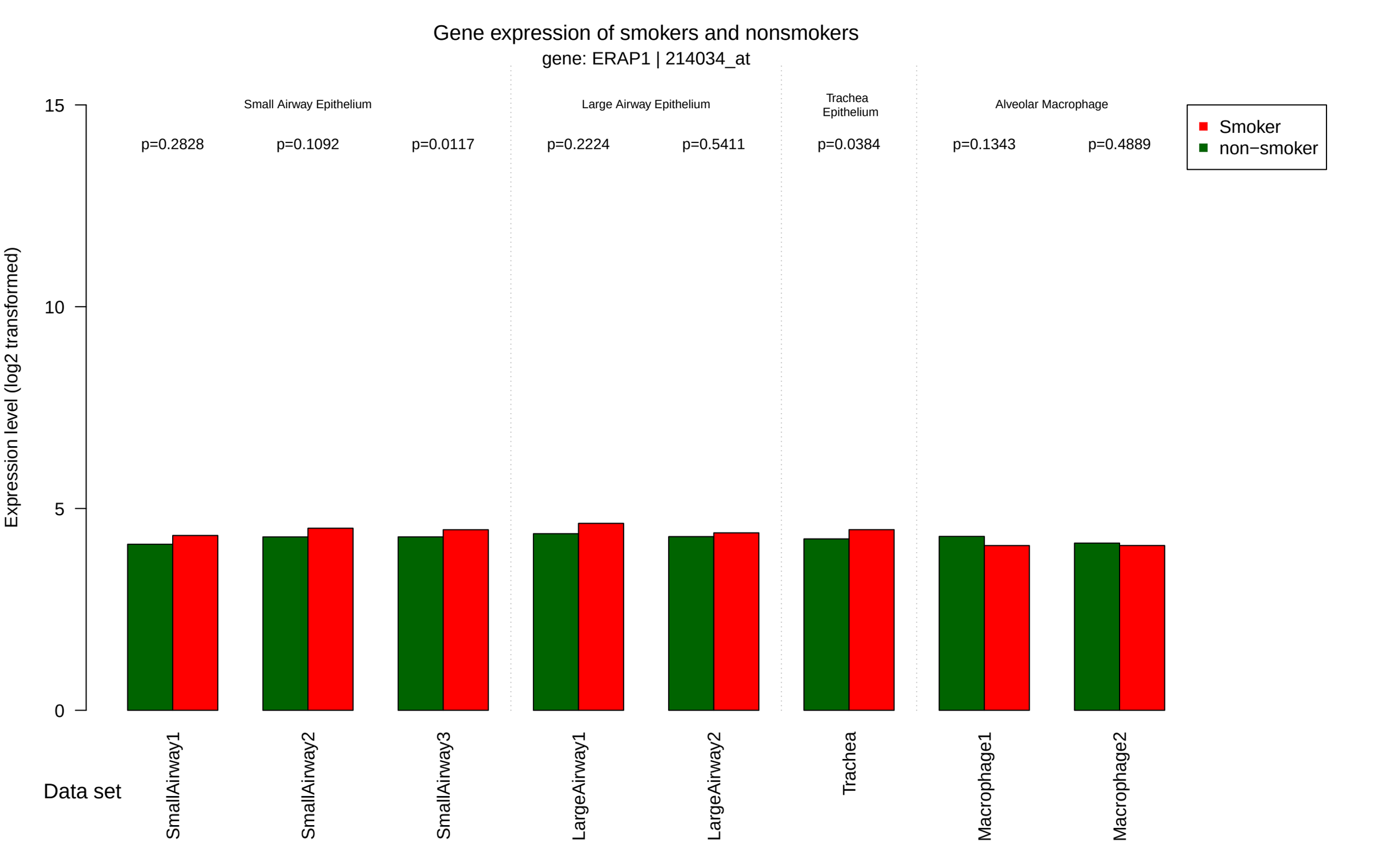
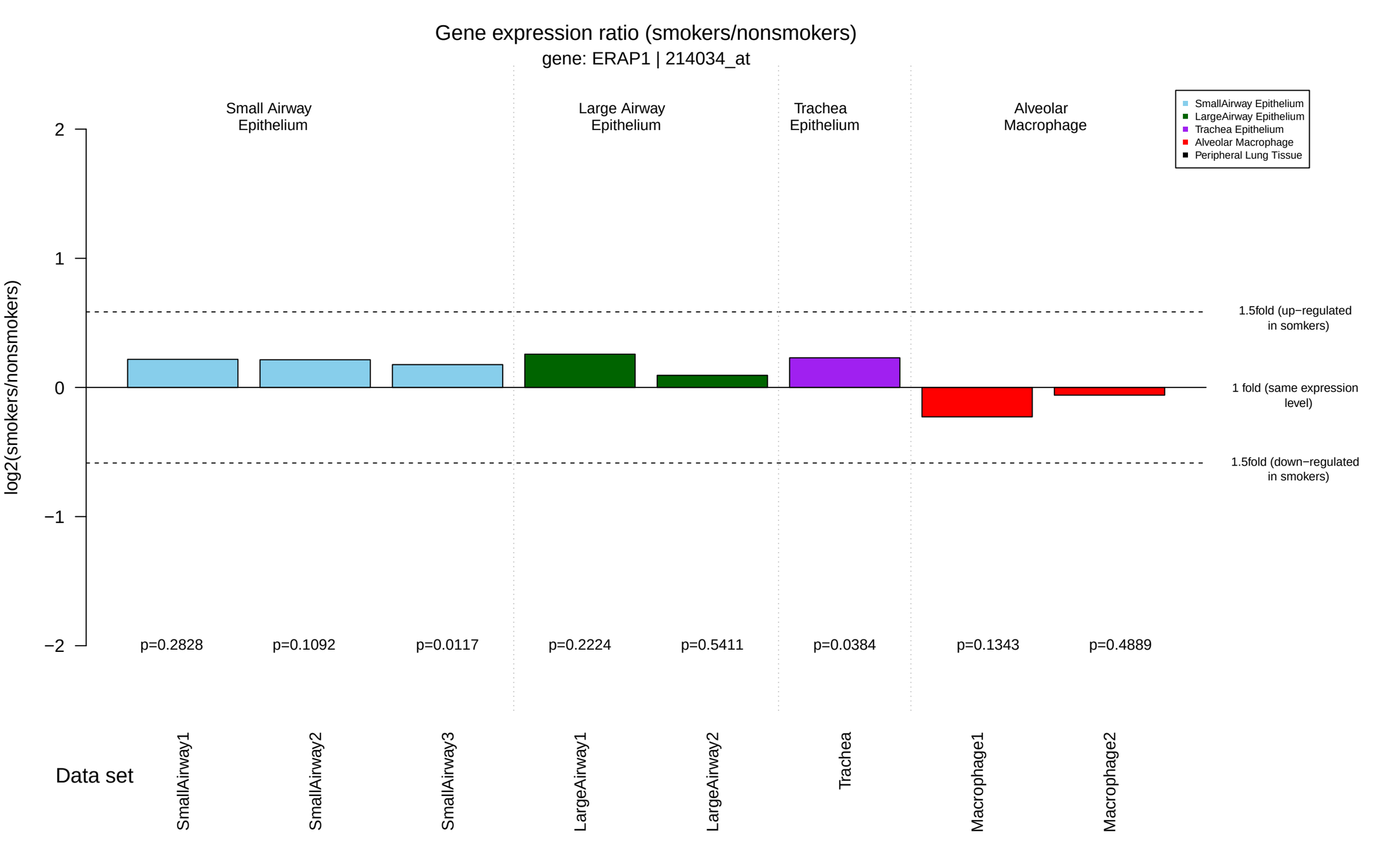
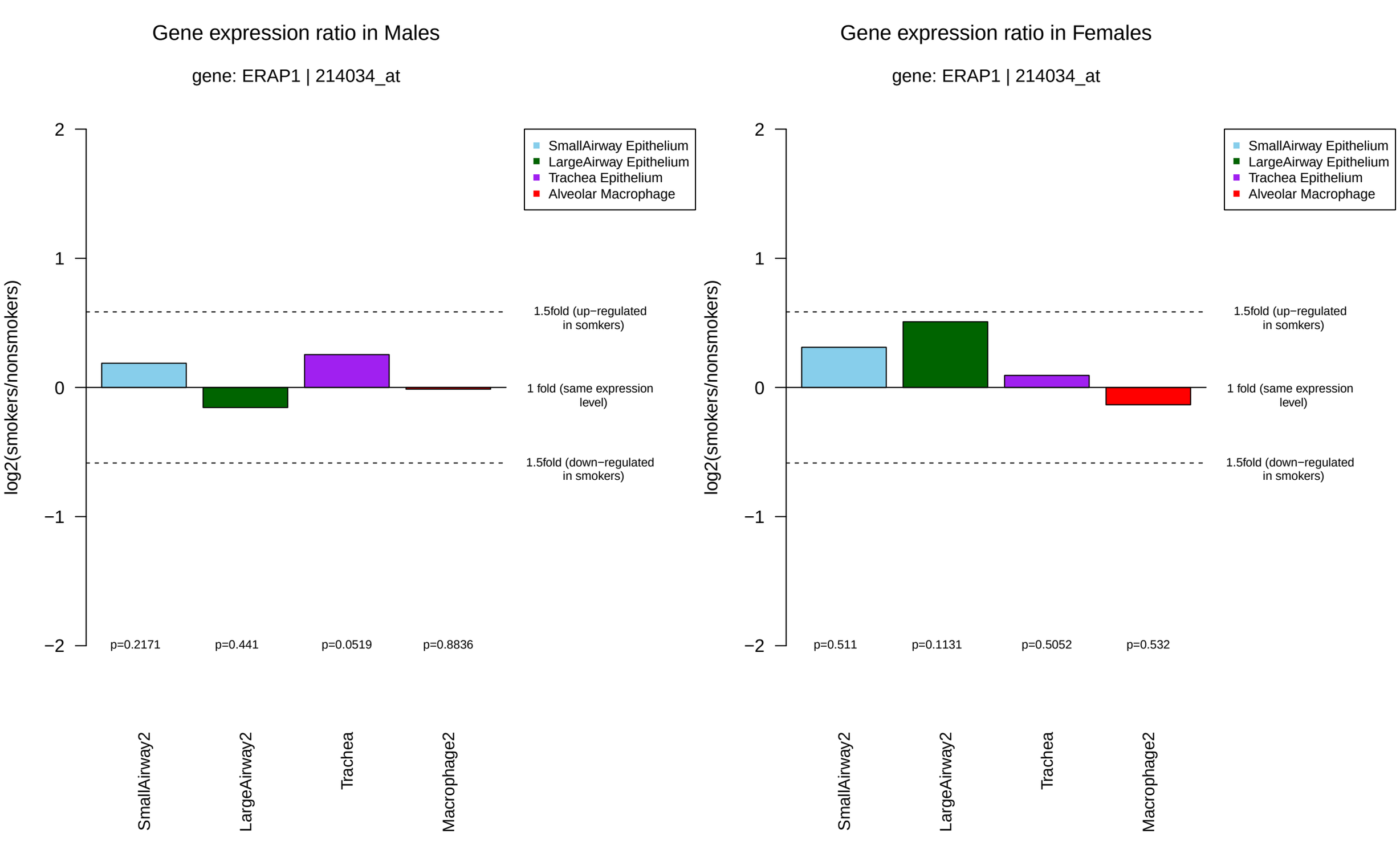
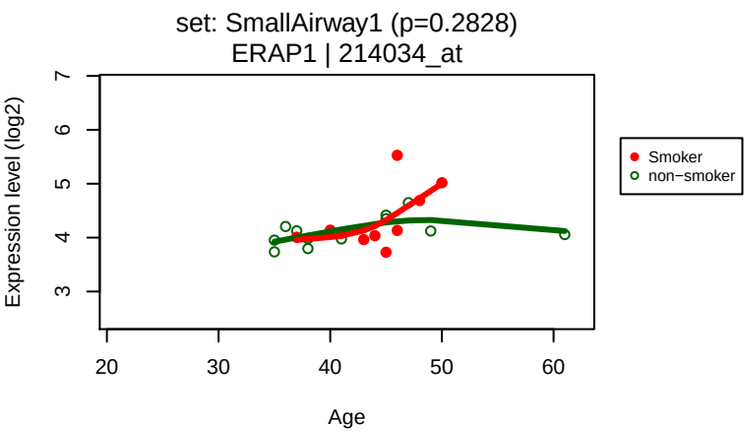
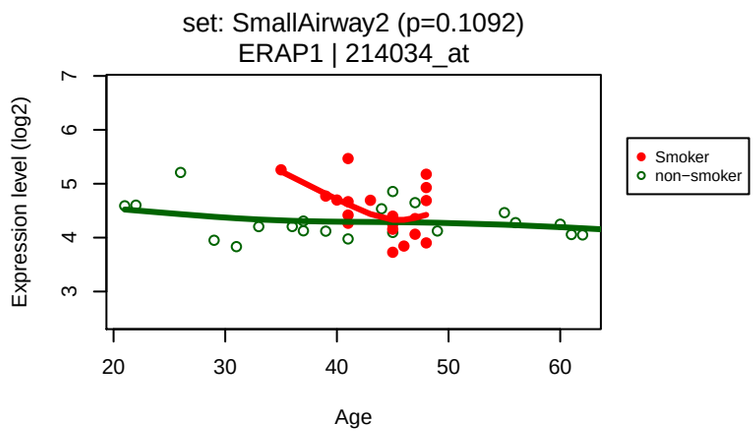
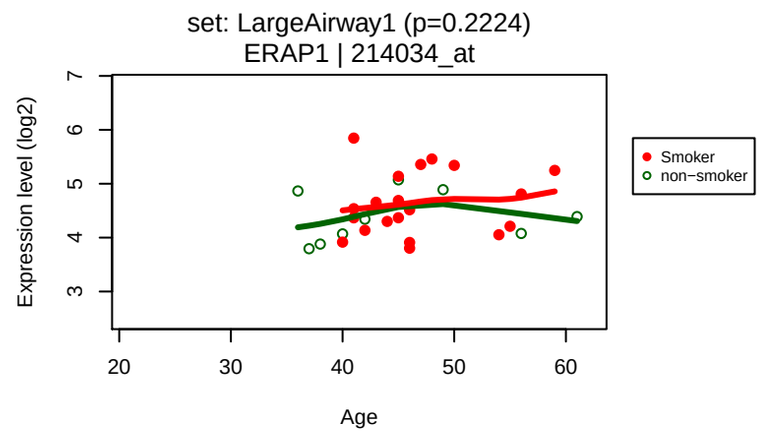

Gene expression of smokers and nonsmokers
gene: ERAP1 | 214034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2828
p=0.1092
p=0.0117
p=0.2224
p=0.5411
p=0.0384
p=0.1343
p=0.4889
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

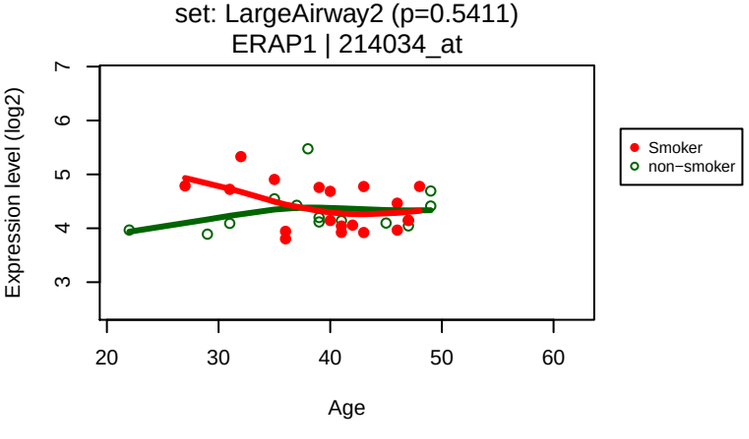

Gene expression ratio (smokers/nonsmokers)
gene: ERAP1 | 214034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2828
p=0.1092
p=0.0117
p=0.2224
p=0.5411
p=0.0384
p=0.1343
p=0.4889
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

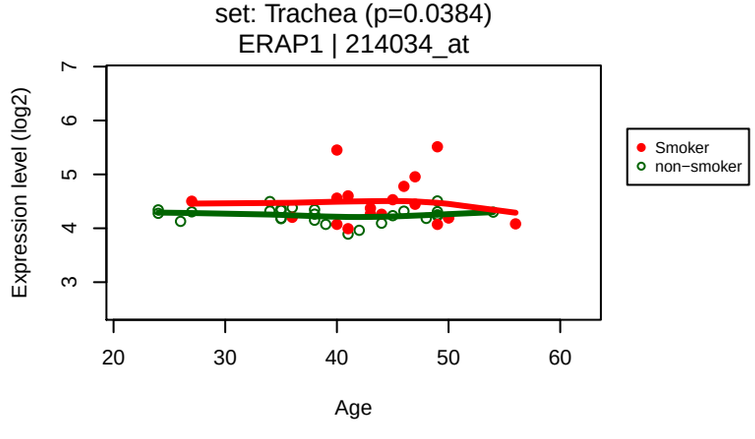

Gene expression ratio in Males
gene: ERAP1 | 214034_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2171
p=0.441
p=0.0519
p=0.8836
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP1 | 214034_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.511
p=0.1131
p=0.5052
p=0.532
SmallAirway2
LargeAirway2
Trachea
Macrophage2

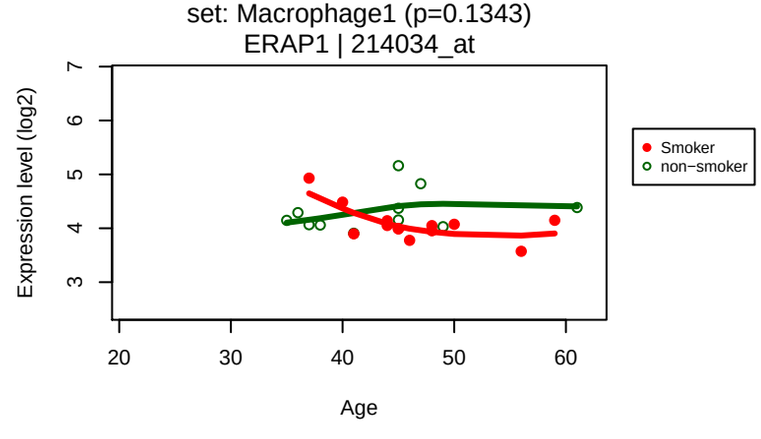

set: SmallAirway1 (p=0.2828)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker

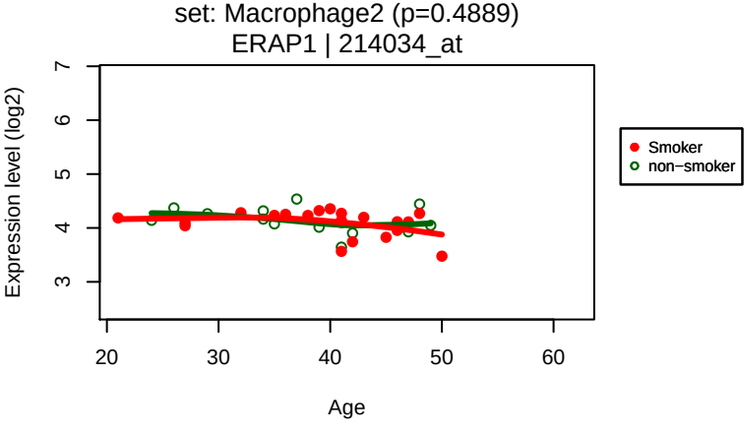

set: SmallAirway2 (p=0.1092)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker

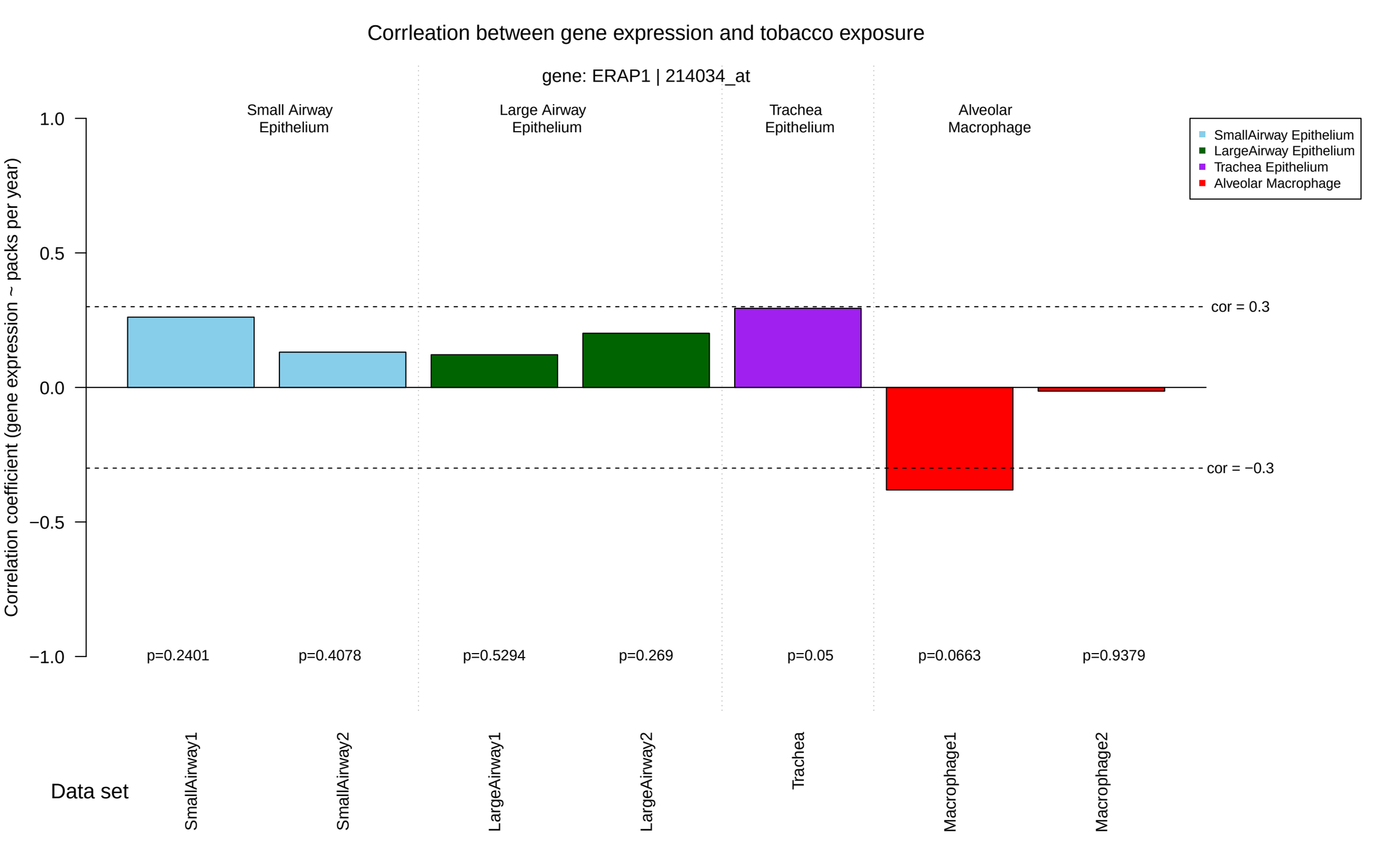

set: LargeAirway1 (p=0.2224)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5411)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0384)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1343)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4889)
ERAP1 | 214034_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP1 | 214034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2401
p=0.4078
p=0.5294
p=0.269
p=0.05
p=0.0663
p=0.9379
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2